

Abstract

In the past several years, the world has managed to transition away from simple automation to independent AI systems. Agentic AI is an agent that can work independently, carrying out all essential plans and implementations without any kind of supervision from a human being. This review has tried to demonstrate the transformative impact that Agentic AI brings to contemporary models of intelligence by means of synthesis of perception, reasoning, and goal. We utilized the phrases Agentic AI, autonomous AI, multi agent systems as keywords in Google Scholar, ScienceDirect, arXiv, and other digital libraries. We have used these 38 main papers published between 2023 and present as part of search process. This paper has taken a deep, close approach to Agentic AI systems, thereby discussing their technical foundations, challenges, applications, and societal impacts. In this paper, we discuss the different aspects of Agentic AI, including the technical side, challenges, applications, and impacts. In this study we found Agentic AI systems are capable of a 34.2 % reduction in the time of task execution, a 7.7 % increase in accuracy, and a 13.6 % improvement in resource utilization. In this survey, we investigate the performance and benchmarking of model architectures, security, privacy, and human-in-the-loop integration; and analyze frameworks for efficient human-AI collaboration. This survey draws on ideas from recent publications, equipping researchers with a holistic view of the state, challenges, and directions for future research necessary to keep Agentic AI safe, aligned, and beneficial to society.

OBJECTIVES

The main goal of this survey is to:

Provide a unified taxonomy for classifying Agentic AI systems by autonomy level and cognitive complexity

Propose a multi-dimensional evaluation framework that goes beyond traditional accuracy metrics

Critically analyze gaps between claimed and actual system capabilities across 8 major frameworks (AutoGPT, LangChain, AutoGen, MetaGPT, CrewAI, LangGraph, Semantic Kernel, Agent Hospital)

Examine security risks, ethical considerations, and human-in-the-loop integration strategies

Methodology

Search Strategy:

Databases: Google Scholar, ScienceDirect, arXiv, IEEE Access

Keywords: Agentic AI, autonomous AI, multi-agent systems, AI memory systems

44 main papers + 39 supplementary survey/review articles (2023–2025)

Over 90% of publications from 2024–2025

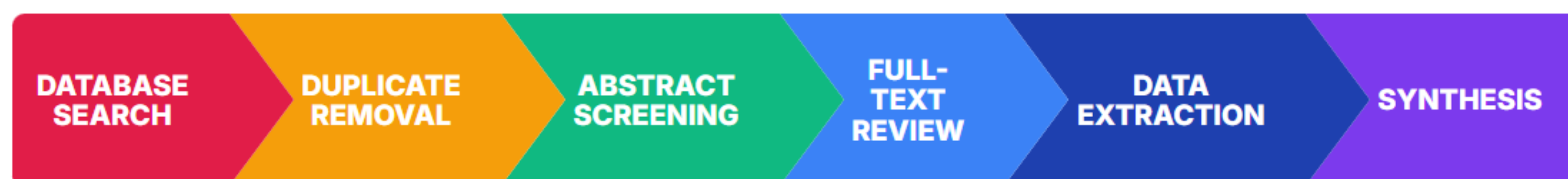
Process:

Removed duplicates, screened abstracts and conclusions for relevance

Included publications analyzing Agentic AI concepts, systems, or issues

Cross-referenced bibliographies for additional sources

Extracted and synthesized information across all selected papers



Proposed Taxonomy

Dimension 1: Autonomy Level

L1 (Assisted): System recommends; humans decide

L2 (Supervised): Autonomous with human approval at critical steps

L3 (Conditional): Autonomous within predefined safety bounds

L4 (Full Autonomous): Self-directed operation, minimal human oversight

Dimension 2: Cognitive Complexity

C1 (Single-Task): Isolated operations, no planning

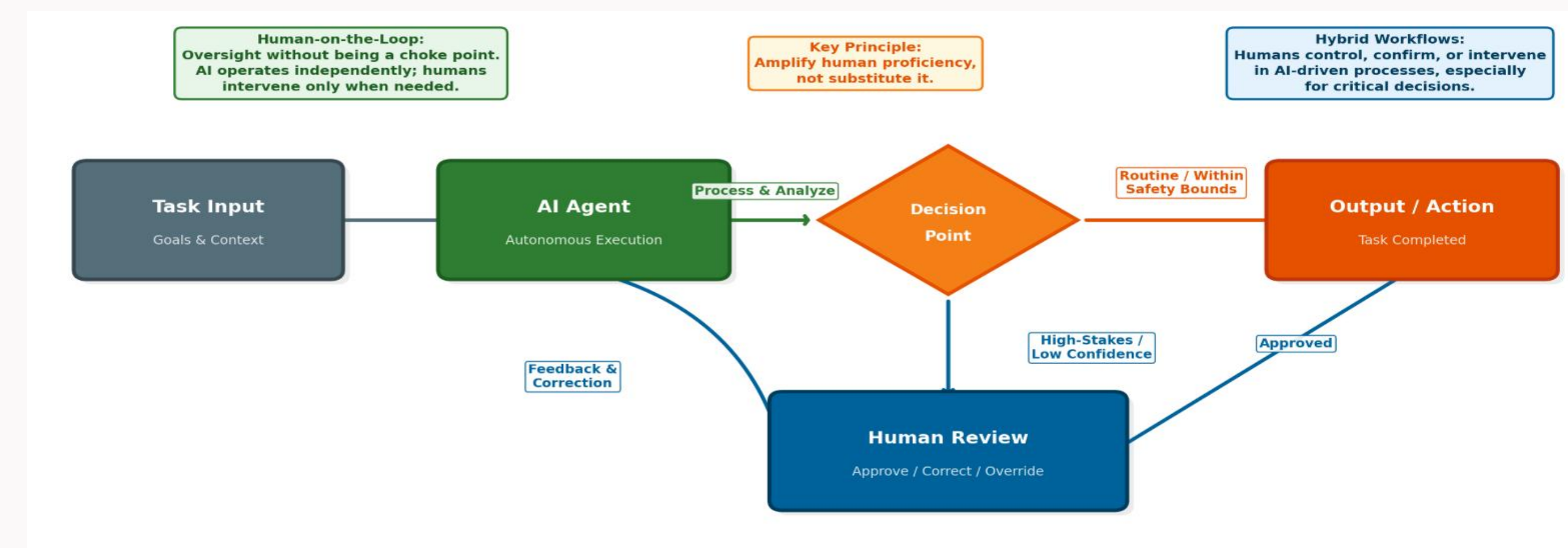
C2 (Multi-Step): Sequential planning with tool integration

C3 (Multi-Agent): Distributed agents with role specialization

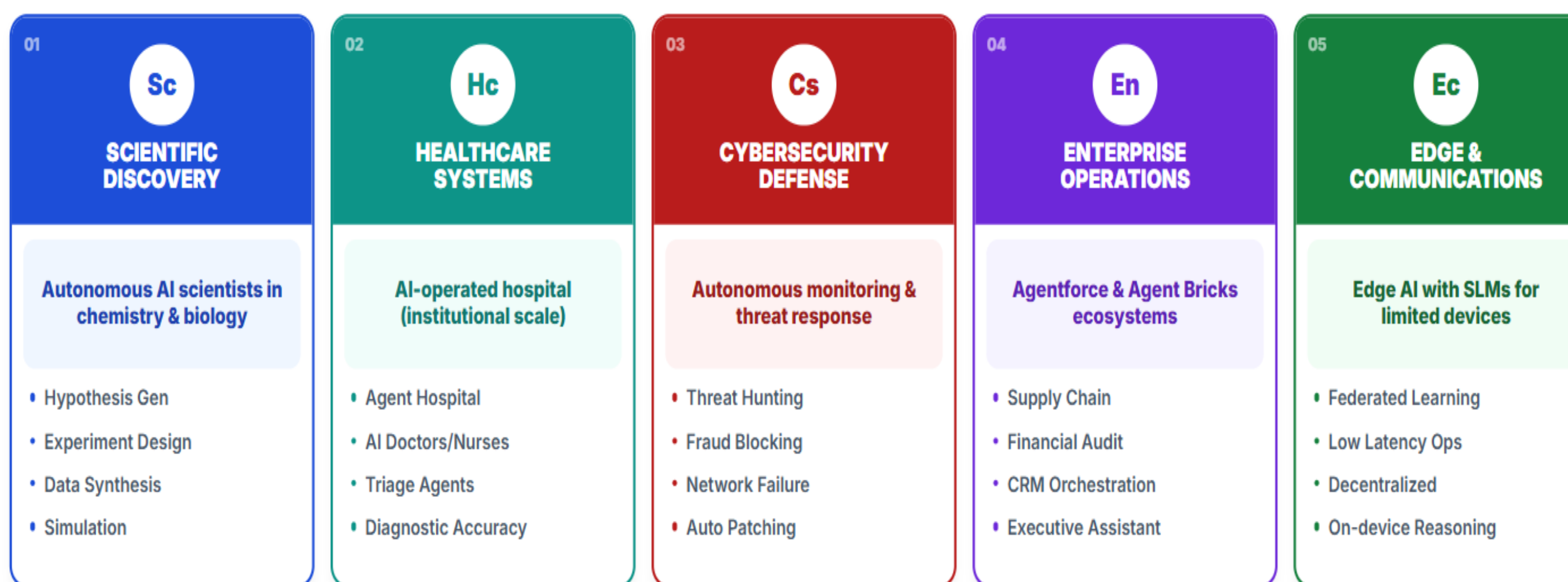
C4 (Self-Evolving): Dynamic goals and autonomous capability expansion

Framework	Autonomy	Complexity	Memory
AutoGPT	L3	C2	LIMITED
LangChain	L2	C2	BASIC
AutoGen	L3	C3	YES
MetaGPT	L3	C3	YES
CrewAI	L2	C3	BASIC
LangGraph	L2	C2	YES
Semantic Kernel	L2	C2	YES
Agent Hospital	L4	C3	YES

Autonomy Level	COGNITIVE COMPLEXITY			
	C1 Single-Task	C2 Multi-Step	C3 Multi-Agent	C4 Self-Evolving
L4: Full Autonomous	–	–	Agent Hospital	–
L3: Conditional	–	AutoGPT	AutoGen MetaGPT	–
L2: Supervised	–	LangChain LangGraph Semantic Kernel	CrewAI	–
L1: Assisted	–	–	–	–

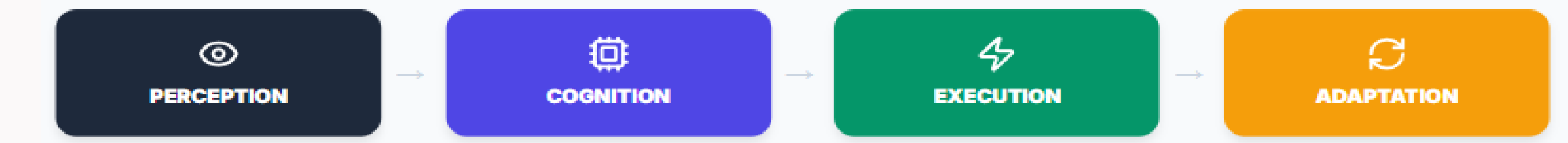


Applications

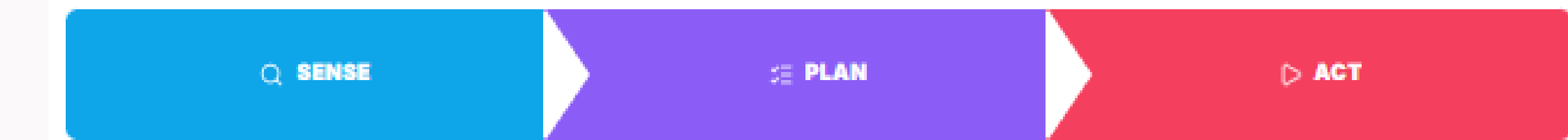


Architectural Paradigms

MODULAR COMPONENT-BASED



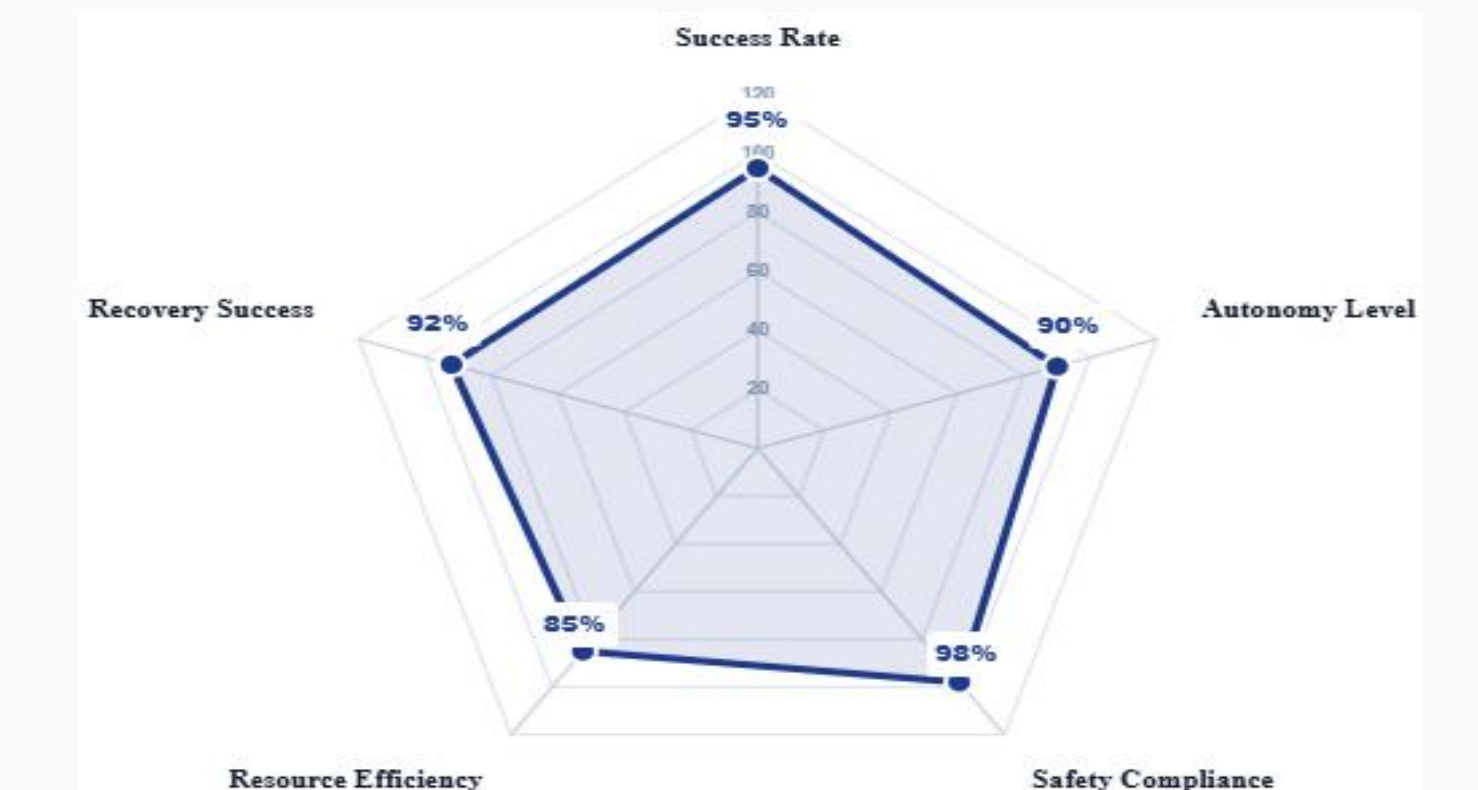
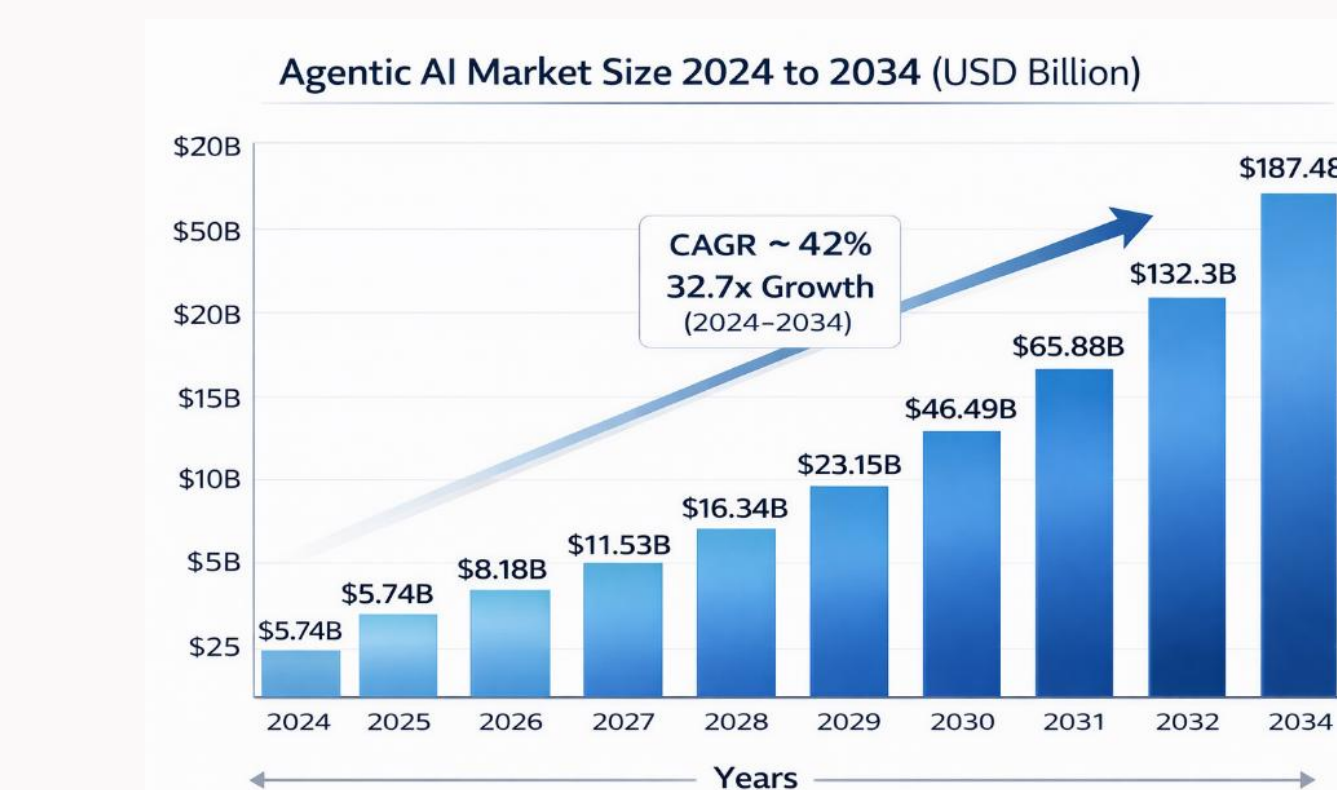
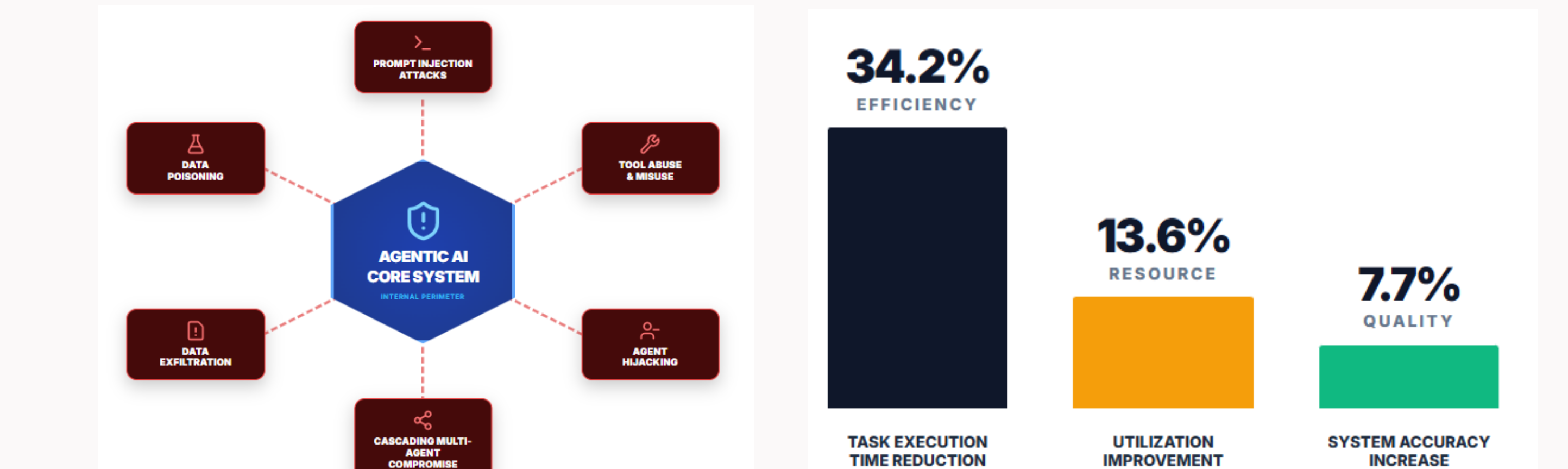
CYCLIC PROCESS (SPA LOOP)



HIERARCHICAL LAYERED



Key Results & Findings



References

- [1] A. K. Pati, "Agentic AI: A Comprehensive Survey," *IEEE Access*, 2025.
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- [3] P. D. Sawant, "Quantitative Analysis of Agentic AI Adoption and Security Concerns," *Journal of Artificial Intelligence Research*, 2025.
- [4] Y. Shavit *et al.*, "Governance and Safety Frameworks for Autonomous AI Agents," 2024.
- [5] M. Gridach *et al.*, "Agentic AI for Scientific Discovery," arXiv preprint, 2025.